

A novel approach of Machine learning application in Astrophysics: Morphological feature wrapping based ensemble method for Galaxy shape classification using GAMA dataset

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Abstract. Numerous supervised machine learning algorithms are used for the automated morphological categorization of galaxies, but there hasn't been a complete evaluation or comparison of these approaches using imaging data. The majority of star galaxy classifiers in use today employ condensed summary data from catalogs, which necessitates careful feature extraction and selection. Machine learning advancements, Based on machine learning, Deep Convolutional Neural Networks (CNN) allow you to automate the recognition of features from data by a machine, reducing the need for qualified human input. Relating to low artificial categorization, significant improvement has been done. Although this is the case, Deep Learning routinely outperforms conventional computers. managing extensive datasets while learning. We analyze the effectiveness of three machine learning algorithms—Support Vector Machines (SVM), Random Forests (RF), and Naive Bayes—in classifying morphological galaxies (NB). With the use of both a set of morphic features generated by image analysis and the raw image pixel data compressed using PCA (Principal Component Analysis) into PCA features, we evaluate the effectiveness of the various machine learning techniques on each feature representation of a galaxy. According to our experiments, RF performed better than SVM and NB. Other than that, our PCA features performed worse than morphic features. Thus, the current mechanism is not extremely scalable. To address these issues, it's critical to create a probabilistic classifier that can scale that is based on source data and uses the least amount of human intervention possible.

Keywords: *Machine Learning, GAMA Dataset, Supervised Classifiers, Galaxy Classification, Morphological Features.*

1 Introduction

Astronomy has always placed a high value on galaxy classification. Galaxies appear in a wide variety of forms, sizes, and hues. As a result, categorizing them helps scientists understand how the cosmos formed and evolved. The categorization of galaxies is based on their form, structure, metallicity, brightness, and other features. There are several strategies to achieve this goal. Galaxy classification is the first step towards better comprehending the physics of galaxies. The availability of several digitized sky studies at various wavelengths has lately transformed astronomy into an exceedingly data-rich science. These pictures are normally categorized by visually examining photographic plates. However, because it requires knowledge and is not an easy task. The evolution of astronomy in the modern period has greatly mirrored technological advancements in computers. Due to the enormous amount of data available to us now and the greatly enhanced efficiency of computer performance, sophisticated computational approaches like machine learning models are now considerably more effective. A large number of seasoned individuals used to classify galaxies by hand long ago, and they would then analyze the findings using a cross-validation procedure. I'm going to expose to a Galaxy Classification Model with Machine Learning in light of this motivation. The SDSS(Sloan Digital Sky Surveys), one of the largest sky surveys, has produced a database of countless millions of galaxies. Since it is hard to manually analyze them, so Galaxy Zoo Project started in 2007 as a crowdsourcing technique to categorize astronomical photographs. The project was successful and finished much more quickly than expected. It is more practical to employ these automated procedures, providing they have great accuracy, due to developments in computer vision and machine learning, particularly Neural Networks, which can complete the operation very instantaneously in Figure 1.

The objective of the study: The study of morphological galaxy classification plays an important role in astronomy: the frequency and spatial distribution of galaxy types provide valuable information for the understanding of galaxy formation and evolution.

Scope of the study: We compare the classification performance of Machine Learning algorithms: Support Vector Machine (SVM), Random Forests (RF), and Decision Tree as applied to morphological galaxy classification. Using both sets of morphic features derived from image analysis and the direct image pixel data compressed through PCA (Principal Component Analysis) into PCA features, we compare the performance of the different Machine Learning algorithms on each feature representation of a galaxy. The experiment shows that Random Forest (RF) performed better than Support Vector Machine (SVM) and Decision Tree. Also, morphic features were more effective than the PCA features.

In line with our earlier findings and the studies from the literature stated above, we discover that the used categorization scheme is not entirely supported, even when taking into account additional catalogs and a different classifier. The Little Blue Spheroids galaxy class, which was recently proposed in [4], interestingly, remains the most prominent, also for the collection of catalogs examined



Fig. 1. (a) Galaxy classification Sensor-based System

in this work, corresponding to our prior work [3]. We list the variables that have the most bearing on the established class divisions. The paper is organized as follows section 1, comprises the Introduction, Section 2 literature review, and research gap, section 3 methodology, and section 4 results and discussion.

Dataset

In the current study, we look at data from five galaxy catalogs (that include features derived from spectroscopic and photometric observations, i.e., measurements of flux intensities in different wavelength bands from the Galaxy AndMass Assembly (GAMA) survey [7] for a sample of 1295 galaxies. Due to the catalogs' ability to give data for multiple groupings of galaxies, our data set consists of the set of galaxies for which a complete set of attributes may be accessed after balancing the relevant classes.

2 Literature review and Research gap findings

Three Machine Learning algorithms SVM, RF, and NB were compared by [1], and they concluded that RF performs better than the SVM algorithm because the features were not linearly separable even when projected to higher dimensions with the RBF kernel. [1], trained SVM, RF, and NB classifiers on training subsets and measured the classification accuracy. They used libSVM, WEKA Random Forest, and [2] NB. They comprised the data set of 119 images with classification labels, which was obtained from Zsolt Frei's galaxy catalog. In [3], Fuentes presented an experimental study of Machine Learning and image analysis for performing automated morphological galaxy classification. They used a neural network and logically weighted regression method. [4], Fuentes developed the method for distinguishing galaxy images into three parts: the image analysis module (IAM), the data compression module (DCM), and the machine learning module (MLM).

They implemented locally weighted regression in MATLAB and used the feed-forward network which was implanted in MATLAB neural network toolbox. They used 10-fold cross-validation for all experiments. They concluded that the algorithms and the ensembles obtained very good results. And their experiment shows that the homogenous ensemble of locally weighted regression obtained better results than the previous report in the literature. And they also got to know that locally weighted regression is faster than artificial neural networks and they also obtain better results. In [6], authors applied decision tree learning algorithms and fuzzy inferencing systems for galaxy morphology classification. They studied The CART, the C4.5, the Random Forest, and fuzzy logic algorithms and developed a reliable classifier to distinguish between spiral galaxies, elliptical galaxies, or star/unknown galactic objects. After comparing the CART, the C4.5, the Random Forest decision tree algorithms, and the output from the Fuzzy inference system, they got the result that Random Forest gave the highest percentages accuracy, especially when 50 trees were used. In [7], presented a star galaxy classification framework that uses deep Convolutional Neural Networks (ConvNets) directly on the reduced, calibrated pixel values. Edward J. Kim and Robert J. Brunner used the data from the Sloan Digital Sky Survey (SDSS) and the Canada-France-Hawaii Telescope Lensing Survey (CFHTLenS) to demonstrate that ConvNets can produce accurate and well-calibrated probabilistic classifications that are competitive with conventional machine learning techniques. They came to the result that for CFHTLenS data set, the network was able to provide a classification that is as accurate as a Random Forest algorithm (TPC), while the probability estimated that ConvNet model appeared to be better calibrated. And when the same network architecture was applied to SDSS data set, the network failed to outperform TPC, but the probabilities were still slightly better calibrated. [8- 10] carried out a comparison between several common machine learning methods for galaxy classification Convolutional Neural Networks (CNN), K-nearest neighbor, logistic regression, Random Forest (RF), Support Vector Machine (SVM), and Neural Networks by using Dark Energy Survey (DES). James Annis showed that CNN is the most successful method of these methods in his study. They used a sample of 2800 galaxies with visual classification from Galaxy Zoo 1, they reached an accuracy of 0.99 for the morphological classification of ellipticals and spirals. They further found that galaxies having a low probability of being either spirals or ellipticals are visually lenticulars, demonstrating that supervised learning can rediscover that this class of galaxy was distinct from both ellipticals and spirals. They also tested how using rotated images with various angles of rotation with 10 deg increments to augment their data influences their classification.

3 Methodology

It doesn't matter if it can be measured or not; a feature is a property of an object that is employed to distinguish (or classify) two unrelated items. To sort arriving fish on a conveyor depending on species, for instance, use optical detection. When an input sample is represented by its features, points in the feature space are utilized as a representation. In the case of a single feature, the feature space has just one dimension. As is well known, pattern recognition seeks to identify the most appropriate decision rule for categorizing the incoming data (classes). Points in one class and points in another are separated by the decision boundary. Decision regions are created by dividing the feature space into decision areas [11].

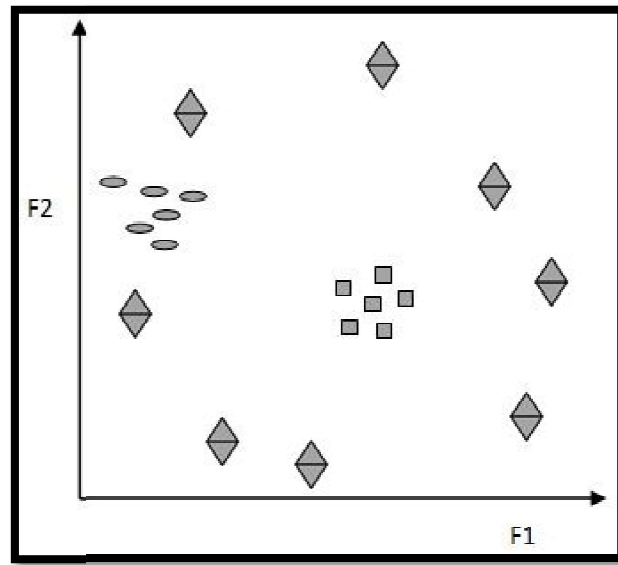


Fig. 2. Sample Points in a 2-Dimensional Feature Space

It is convenient to express pattern classification issues in terms of discriminant functions since these functions may be readily computed from data. When the decision boundaries are computed directly, the technique is commonly referred to as discriminative modeling. The decision to choose a particular functional form whose parameters have been altered by a training strategy or prior knowledge of the patterns to be classified may be the basis for the discriminant function selection. In the literature, a variety of discriminant function types have been studied, ranging in complexity from the simple linear discriminant function (where g is a linear combination of the x_i) to sophisticated multi-parameter.

nonlinear functions like the multilayer perceptron. All model parameters, such as class priors and feature probability distributions, may be estimated using the relative frequencies of the training set. The probabilities are predicted with the highest degree of accuracy [12-14]. One can estimate the class probability from the training set, using the formula (prior for a specific class) = (number of samples in the class) / (total number of samples), or one can assume equiprobable classes (i.e., priors = $1 / (\text{number of classes})$). A distributional assumption must be made or nonparametric models must be built for the features in the training set to estimate the parameters for a feature's distribution. Working with continuous data frequently involves assuming that each class's associated continuous values are distributed Gaussian-style. Another usual approach for dealing with continuous variables is to bin data and discretize the data. The distribution strategy is usually the better choice if the training data is sparse or the precise distribution of the data is known. The discretization strategy frequently outperforms other approaches in bigger training data sets because of its flexibility in adjusting to the distribution of the training data. Given that naive Bayes is usually used when a large amount of data is available, the discretization strategy is frequently preferred over the distribution method (as more computationally intensive models can normally yield greater accuracy). In figure 3, the proposed model is given [15-17].

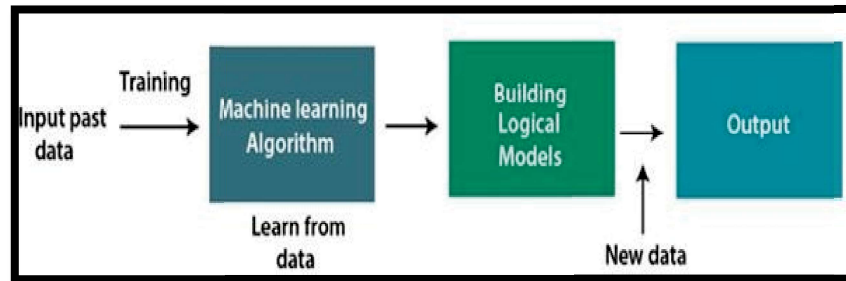


Fig. 3. proposed model

Deep Learning: A neural network comprising three or more layers is a neural network, and Deep learning is a component of machine learning. These neural networks make an effort to simulate how the human brain functions, however, they fall well short of being able to match it, allowing it to “learn” from huge amounts of data. A neural network only with one layer can still create approximations, but more hidden layers could even help to improve and enhance the accuracy as shown in Figure 4 [18].

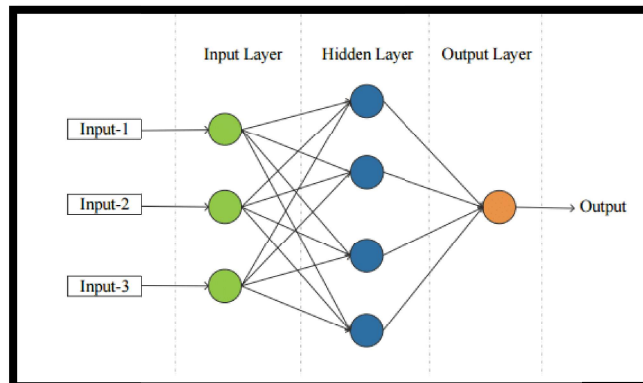


Fig. 4. Deep learning algorithms

Logistic Regression: A machine learning technique called regression examines the outcomes of a variable that can only be answered yes or no. It belongs to the supervised learning methodology. As a result, it has a distinct and category value. The category of the variable that is more typical can be easily identified. Logistic regression and observations of various types of data are the two categories into which it can be separated. Values from the memorial and binomial regression are used to define logistic functions.

Support Vector Machine: The SVM, which is used to address respectively classification and as well as regression problems, is one of the most used supervised learning techniques. However, machine learning classification problems make up the majority of its use. The SVM algorithm's objective is to establish the decision boundary or best line that can divide n-dimensional space into classes, allowing us to quickly classify fresh data points in the future. A hyperplane is a name that is given to this optimal decision boundary.

Random Forest: Machine Learning popular method Random Forest is a part of the supervised learning approach. It is applied to both Regression and Classification problems in Machine Learning (ML). This is based upon the concept called ensemble methods, which is the technique of integrating different classifiers to handle a complicated issue and enhance the effectiveness of the model. Random Forest, as the name implies, is indeed a classifier that uses a collection of decision trees upon different subsets of the provided dataset and averages them to increase the dataset's predictive accuracy [15].

Decision Tree: A decision Tree is a supervised machine learning method that can be applied to classification and regression issues; however, it is most frequently used to address classification issues. It consists of a tree-structured classifier, with internal nodes denoting dataset features, branches denoting decision rules, and each leaf node denoting the classification result. Two nodes- the Decision node and the Leaf node make up a decision tree. A choice is made using a decision node, which has several branches, whereas a Leaf node is the result of that decision and does not have any additional branches. Based on the characteristics of the available dataset, decisions or analyses are performed as shown in Figure 5.

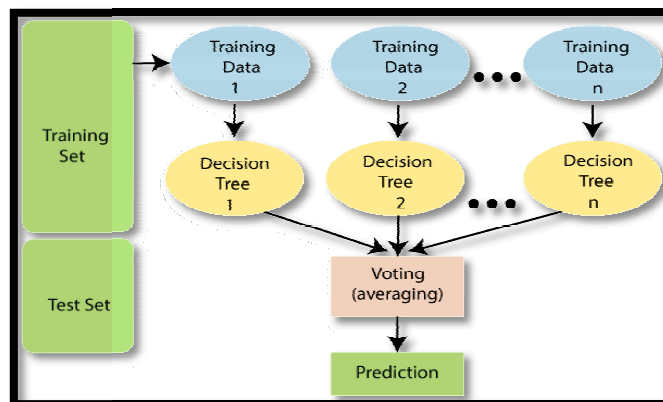


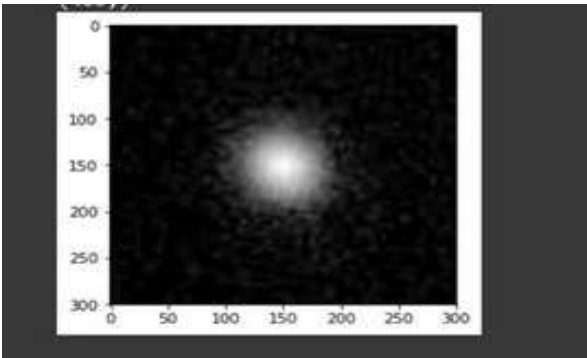
Fig.5. Decision tree and Random forest algorithm

4 Results

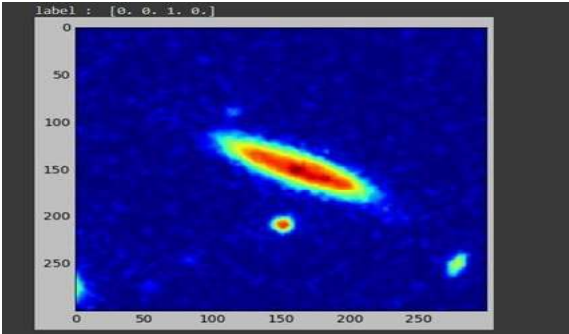
The confusion matrix for the classification based on the Gauss Fit Simple Catalogue shows the highest classification accuracy of 86 per. for the voting classifier. In Class 3, accuracy drops to 47 per. This is partly due to more categorization overlap; 31 % of class 1 sample are also classified as class 3 samples, while 31 % of class 3 samples are classified as class 1 samples. Class 1 and class 3 overlap with LBS and class 7 overlap are both growing (12 per, but they are still very small shown in Figure 6 (A)). Class 1 and class 5 overlap is also growing (12 per. To provide readers with a better understanding of the nature of the utilized visual-based categorization approach, particularly concerning physical variables, we have highlighted the relevances of the catalog components for the analyzed class differences in Figure 6 (b). It should be emphasized that relevant interpretation requires taking into account the characteristics of the data sample, such as correlations and classification performance shown in Figures 6 (c) and (d). The binary classifier system that was created was based on the area under the receiver operating characteristic curve (ROC). The ROC curve is produced by contrasting the classifier's True Positive Rate and False Positive Rate. The metric, which goes from 0 to 1, is inversely proportional to how accurate the forecast was. This statistic essentially assists in our understanding of how the classifier performs when the threshold is changed shown in Table 1 using a comparison with existing methods.

5 Conclusion and future scope

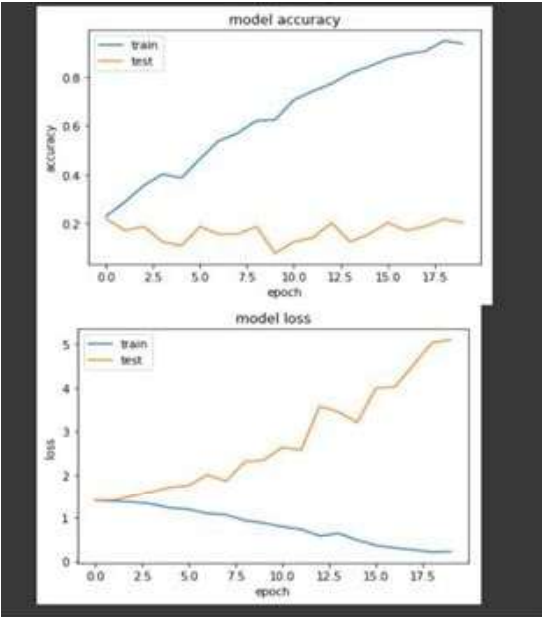
The implemented system takes predefined constant image data set as input but it can be improved or updated to get a defined data set. The implemented system provides the result of multiple classified types of galaxies present in the given data set on which it has been trained on. After carrying out the implantation and contrasting the different Machine Learning algorithms, we can say that Random Forest is the best and has a high accuracy rate for categorizing galaxies into ellipticals and spirals. The accuracy of the estimated morphic features was influenced by several factors. The clump of stars and dust that ringed the galactic nucleus caused erroneous margins, making it impossible to determine the galaxy's edge. Future research will employ more training data to improve classification accuracy and will integrate morphic qualities with photometric information from other spectra. The bar-to-bulge ratio and integrated curvature morphic properties may also help to increase the seven-category classification's precision. It is recommended to use a proper and good data set that will yield proper and accurate results. Uses of good quality image data sets will be valued.



(a)



(b)



(c)

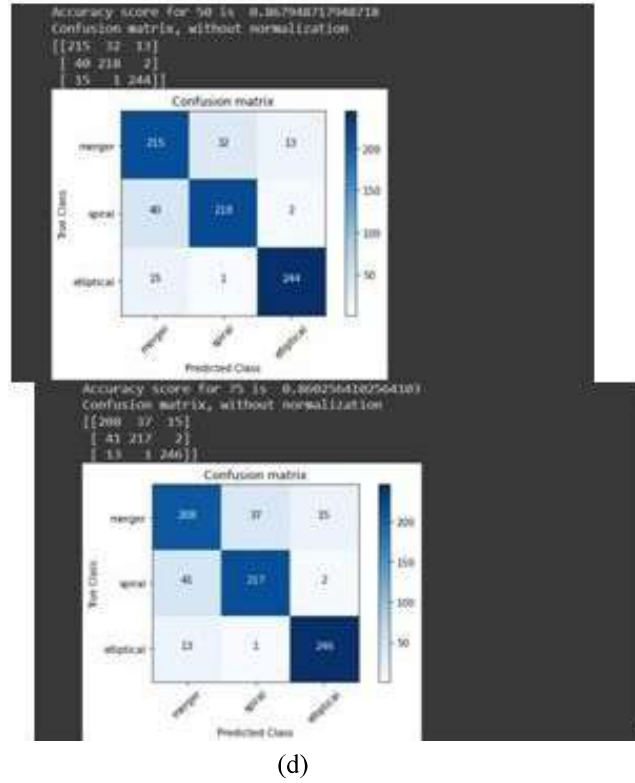


Fig. 6. (A) Sample data (B) morphological image
(C) Confusion matrix (D) ROC and AUC curve

Table 1. Comparison of the existing method with proposed one

Paper No.	Data Set	Methods	Accuracy
[2]	GAMA	Used Naïve Bayes classifier, the rule induction algorithm C4.5 and random forest.	91.64%
[8]	GAMA	Used weighted KNN along with Fischer score	90%
[13]	GAMA	CART, C4.5, random forest, and fuzzy algorithms were used, of which random forest(50 trees) proved most accurate.	91.1%
Proposed one	GAMA	Morphological feature selection and reduction for polynomial time/ computational complexity reduction and classification through ensemble method.	93.55 %

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